Temporal Fractal Coupling of Volatility and Trading Volume: A Robust Within-Firm Pattern and No Firm-Specific Forecast Power in S&P 500 Stocks, 2015–2026

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Abstract

Do the fractal scaling properties of price volatility and trading volume co-evolve, and does their co-movement carry predictive information about future market conditions? I apply detrended fluctuation analysis to 488 current S&P 500 constituents over 2015–2026, computing rolling Hurst exponents for absolute returns ($H^{(i)}_{|r|,t}$) and log-volume ($H^{(i)}_{v,t}$) in aligned windows, and introduce the Coupling Intensity Index (CII), a rolling Pearson correlation that measures their temporal co-movement. Within-firm temporal coupling is strong and positive: mean $r = 0.531$ across the 488-firm panel, with 92.7% of firms exhibiting positive coupling and 80.4% above $r = 0.3$. The pattern holds across all eleven GICS sectors. The cross-sectional analogue at $G = 488$ shows a small but statistically detectable positive correlation between full-sample $H_{|r|}$ and H_v ($r = 0.18$, $p < 0.001$); within-firm temporal coupling is roughly threefold stronger and structurally different. Coupling heterogeneity rises sharply in high-VIX regimes (Mann-Whitney high vs. low $p = 0.0019$, with the upper-quartile distribution exhibiting both higher medians and substantially greater dispersion), but mean coupling does not uniformly intensify with stress. The predictive question returns a clean null. Under inference robust to firm-clustering, two-way clustering by firm and time (Cameron et al., 2011), Bell–McCaffrey CR2 with Satterthwaite degrees of freedom (Imbens and Kolesár, 2016), and the wild cluster restricted bootstrap (MacKinnon

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and Webb, 2017), CII has no firm-conditional incremental predictive content for any of the ten forward outcomes tested in this paper, including standard dollar-volume Amihud (Amihud, 2002) illiquidity, the Corwin–Schultz spread (Corwin and Schultz, 2012), the Roll spread (Roll, 1984), realised volatility, max drawdown, abnormal turnover, vol-of-vol, log Amihud, ΔAmihud, and forward Corwin–Schultz / Roll spreads. This firm-conditional null is the primary H4 finding, and it is stable across every inference method that conditions on the firm dimension where the panel's dependence is concentrated. A subset of inference methods that condition on the time dimension instead (HC1, time-clustered standard errors, Driscoll–Kraay HAC) return significant focal-only coefficients, but this significance is not robust to the inference choice: those methods do not condition on the firm dimension, and the firm-conditional reading is therefore the conservative and primary one. Every focal-only signal in the battery is in any case fully absorbed by nine standard benchmarks in the combined horse-race specification. An apparently strong predictive result against a non-standard share-volume illiquidity proxy dissipates once both proper Amihud convention and dependence-robust inference are imposed; I document and dissect this as a price-level confound. The contribution is therefore a strong and replicable within-firm fractal coupling phenomenon, a firm-conditional predictive null that holds across the modern cluster-robust toolkit, and three methodological observations for the wider econophysics-meets-finance literature: distinguish share-volume from dollar-volume liquidity proxies, be alert to the sensitivity of a result to the inference dimension when residual dependence is strong on both dimensions, and treat the Bell–McCaffrey Satterthwaite df as a routine diagnostic rather than assuming it scales with cluster count.

Keywords: Hurst exponent, Detrended fluctuation analysis, Price-volume relationship, Long memory, Coupling Intensity Index, Amihud illiquidity, Cluster-robust inference, Wild cluster bootstrap, Methodological caveat

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1. Introduction

Financial price volatility and trading volume each exhibit long-range dependence.[2] Absolute returns display Hurst exponents of approximately 0.7–

[2] This paper builds on an earlier MSc project (Malempati Hari, 2013) in which I examined whether Hurst-based fractal characteristics of stock-price series were associated with

0.8 (Ding et al., 1993; Bollerslev and Jubinski, 1999), while volume persists even more strongly at $H \approx 0.7$–0.9 (Lobato and Velasco, 2000). Whether these two fractal regimes are *coupled* (whether they co-evolve over time, and whether that co-evolution carries information a financial economist would care about) has not been systematically investigated.

This paper addresses both questions. I distinguish between *static* coupling (the cross-sectional correlation of Hurst exponents across firms) and *temporal* coupling (the within-firm co-movement of rolling Hurst exponents over time), and I construct a Coupling Intensity Index (CII) that measures the strength of temporal coupling as a time-varying quantity. I then test four hypotheses:

- **H1** *Static coupling:* Firms with more persistent volume also have more persistent volatility in the cross-section.
- **H2** *Temporal coupling:* Rolling Hurst exponents for $|r_t|$ and $\log V_t$ co-move positively within firms over time.
- **H3** *Regime dependence:* Temporal coupling intensifies during high-uncertainty regimes (elevated VIX).
- **H4** *Predictive content:* CII has incremental predictive content for standard forward liquidity and volatility measures.

The results are as follows. H1 is reframed rather than rejected: the cross-sectional Pearson correlation between full-sample $H_{|r|}$ and H_v at $G = 488$ is small but statistically detectable ($r = 0.18$, $p < 0.001$). The original v1.0 50-firm sample returned a null on the same construction; the broader panel surfaces a modest cross-sectional dependence that the smaller sample lacked the power to detect. H2 is strongly supported at the full S&P 500 scale and is roughly threefold stronger than the cross-sectional analogue: mean within-firm temporal coupling is $r = 0.531$, with 92.7% of the 488 panel firms exhibiting positive coupling and 80.4% above $r = 0.3$; the pattern is

average trading volume in the cross section. That earlier work found a weak but statistically significant static relationship. The present study extends that line of inquiry by shifting from static association to temporal co-evolution, asking whether the persistence structures of volatility and volume move together through time and whether that coupling predicts future market illiquidity.

universal across all eleven GICS sectors. The interesting empirical object remains the within-firm temporal coupling, not the modest cross-sectional level dependence. H3 is confirmed in a sharper form than its original 50-firm version: a Mann-Whitney test for the high-VIX vs. low-VIX coupling distribution rejects the null at $p = 0.0019$, but the rejection is driven by distributional shape rather than mean shift. The high-VIX regime exhibits a higher median of within-firm coupling and substantially greater cross-firm dispersion; mean coupling does not uniformly intensify with stress.

H4 is rejected: CII has no firm-conditional predictive content for forward illiquidity. Under every inference method that conditions on the firm dimension where the panel's dependence is concentrated (firm-clustered, two-way clustered with the Cameron–Gelbach–Miller eigenvalue PSD adjustment, Bell–McCaffrey CR2, and the wild cluster restricted bootstrap), the CII coefficient is statistically null for forward Amihud illiquidity computed in the standard dollar-volume convention of Amihud (2002), and for realised volatility, the Corwin–Schultz spread, the Roll spread, the volatility of realised volatility, the maximum drawdown, abnormal turnover, log Amihud, and ΔAmihud. This firm-conditional null is the robust, primary result. A subset of inference methods that condition on the time dimension instead (HC1, time-clustered standard errors, Driscoll–Kraay HAC) return significant focal-only coefficients for several proxies, but this significance is not robust to the inference choice: those methods do not condition on the firm dimension, so the conservative and primary reading is the firm-conditional null. The horse race against nine standard liquidity benchmarks reinforces the same conclusion: every focal-only signal in the battery is fully absorbed in the combined specification (the strongest case is the forward Corwin–Schultz spread, where focal two-way $t = +3.64$ collapses to combined $t = +0.07$). CII adds no firm-specific predictive content beyond what the standard toolkit already captures.

A specification that uses share volume rather than dollar volume in the Amihud denominator does produce a statistically significant CII coefficient at $G = 50$ and survives the wild cluster restricted bootstrap. That signal disappears once dollar volume replaces share volume in the denominator, as standard practice requires. Section 5.5.5 dissects the divergence as a price-level confound: the share-volume Amihud co-moves with within-firm price-level drift, and CII correlates with that drift component rather than with price impact per dollar traded. The fix is in the dependent-variable definition rather than in the standard errors, and modern cluster-robust inference (including the wild cluster bootstrap) does not detect it.

This combination of results is sharper than the broad predictive thesis the paper originally set out to test. Prior work on price-volume fractal relationships (Podobnik et al., 2009) and common long-memory components (Bollerslev and Jubinski, 1999) has established that volatility and volume share structural features. The present paper shows that the within-firm coupling of those structural features (a) exists in a strong and quantitatively measurable form across the full S&P 500, (b) is regime-dependent in a heterogeneity-amplifying rather than mean-amplifying sense, and (c) carries no firm-conditional predictive content for standard market-quality outcomes once the firm dimension of dependence is properly modelled. The empirical claim is narrower than the original H4 but is robust: the firm-conditional null holds across the full modern cluster-robust toolkit. The focal-only significance that surfaces under time-conditional inference does not condition on the firm dimension where the dependence is concentrated, so it is best read as inference-dimension sensitivity that motivates caution rather than as a separate finding. The paper's robust methodological contribution is the share-volume vs. dollar-volume Amihud confound: an apparently strong predictive result against a non-standard share-volume illiquidity proxy is driven by a price-level component of the denominator and dissolves once the standard convention is imposed.

The conceptual distinction that organises the paper is between first-order and second-order fractal properties. Detrended cross-correlation analysis (DCCA; Podobnik and Stanley, 2008) measures whether price and volume fluctuations are correlated at different scales, a first-order property. The rolling Hurst approach measures whether the *scaling exponents themselves* (which summarise how memory decays) evolve in tandem. This is a second-order property: a statement about the dynamics of complexity rather than the dynamics of levels.

2. Literature

2.1. Long Memory in Returns and Volatility

Mandelbrot (1963) first demonstrated that financial prices exhibit non-Gaussian, scale-invariant properties. The Hurst exponent H, originally developed for hydrological applications (Hurst, 1951) and formalised through fractional Brownian motion (Mandelbrot and Van Ness, 1968), provides the standard measure of long-range dependence. Cont (2001) identified the slow decay of autocorrelation in absolute returns as among the most robust stylised facts in finance. Ding et al. (1993) showed that this long memory is strongest for $|r_t|^d$ at $d \approx 1$, persisting over hundreds of lags. Lo (1991) cautioned that naive R/S analysis can overstate long memory when short-range dependence is present, motivating the use of detrended fluctuation analysis (Peng et al., 1994). Finite-sample properties and heavy-tail biases have been characterised by Weron (2002) and Barunik and Kristoufek (2010). The HAR-RV model of Corsi (2009) provides an influential parsimonious framework for realised volatility forecasting, exploiting the heterogeneous autoregressive structure of realised measures.

2.2. Long Memory in Trading Volume

Lobato and Velasco (2000) provided rigorous evidence for fractional integration in volume, estimating $H \approx 0.7$–0.9 across NYSE equities. This finding is even more robust than the volatility long-memory result and has been replicated across markets and frequencies. Plerou et al. (2003) documented power-law correlations between price changes and volume, while Ray and Tsay (2000) confirmed long-range dependence in daily stock volatilities. Chordia et al. (2001) showed that trading activity predicts expected stock returns through its relationship to liquidity provision.

2.3. Liquidity, Volume, and Market Microstructure

The Amihud illiquidity ratio (Amihud, 2002) has become a standard measure of price impact per unit of trading volume, capturing the adverse-selection component of illiquidity. Kyle (1985) provided the theoretical foundation for relating information asymmetry to market depth. Brunnermeier and Pedersen (2009) formalised the feedback loop between volatility and liquidity: when volatility rises, margins tighten, forcing deleveraging that further reduces liquidity. This volatility-liquidity spiral provides a theoretical channel through which fractal coupling between volatility persistence and volume persistence could predict liquidity deterioration.

2.4. Joint Dynamics: The MDH and Beyond

The Mixture of Distributions Hypothesis (Clark, 1973; Tauchen and Pitts, 1983; Epps and Epps, 1976) posits a latent information arrival process that simultaneously governs volatility and volume. Andersen (1996) recast the

MDH within a stochastic volatility framework, showing time-varying information flow generates the observed co-clustering. Bollerslev and Jubinski (1999) identified a common long-memory component shared by volatility and volume using ARFIMA methods. Podobnik et al. (2009) applied DCCA to measure scale-dependent cross-correlations between price changes and volume changes. Karpoff (1987) and Gallant et al. (1992) established the broader empirical price-volume relationship. Kristoufek (2015) reviewed the methodological challenges and open problems in measuring power-law cross-correlations, including bias and finite-sample issues that motivate the rolling-window approach adopted here.

Several studies have computed time-varying Hurst exponents for individual series. Cajueiro and Tabak (2004) examined emerging-market efficiency. Alvarez-Ramirez et al. (2008) characterised U.S. stock markets through erratic dynamics of the rolling Hurst exponent estimated by DFA. Grech and Mazur (2004) and Grech and Pamuła (2008) used the local Hurst exponent to study crash dynamics on the Polish stock exchange. Garcin (2017) developed a variational-smoothing estimator for time-dependent Hurst exponents and applied it to foreign-exchange rate forecasting, providing the closest existing methodological cousin to the rolling-window approach adopted here. Antoniades et al. (2021) introduced a visual warning tool based on time-dependent generalised Hurst exponents H_q computed in overlapping windows, which is the broader framework into which the rolling DFA construction used in this paper falls as a special case at $q = 2$. Vogl (2023) applied wavelet-denoised rolling Hurst exponents to S&P 500 returns in a nonlinear-dynamics framework. Brouty and Garcin (2024) linked entropy-based market information to the Hurst exponent under fractional Brownian motion and its Lamperti-stationary extension, providing a complementary information-theoretic framing to the present coupling-based approach. Garcin and Grasselli (2022) studied the rough-volatility dilemma across long and short time scales using a large dataset of stock indices and FX rates, documenting finite-sample bias issues in Hurst estimation that motivate the multi-estimator robustness check we report in Section 6. None of these studies, however, examines whether Hurst exponents computed from *two distinct series* co-move through time within a firm, nor whether such co-movement carries predictive content for standard market-quality outcomes. This is the gap the present study addresses.

3. Methodology

3.1. Detrended Fluctuation Analysis

For a time series $\{x_t\}_{t=1}^{N}$, the cumulative deviation profile is $Y(k) = \sum_{t=1}^{k}(x_t - \bar{x})$. The profile is divided into non-overlapping windows of length s, each detrended by a fitted polynomial of order m (here $m = 1$, i.e., DFA-1). The fluctuation function is:

$$F(s) = \left[\frac{1}{N_s}\sum_{\nu=1}^{N_s}\frac{1}{s}\sum_{k=1}^{s}\Big(Y((\nu-1)s+k) - \tilde{Y}_\nu(k)\Big)^2\right]^{1/2}. \tag{1}$$

For a long-range correlated process, $F(s) \sim s^H$. The Hurst exponent H is estimated as the ordinary least squares slope of $\log F(s)$ versus $\log s$. The scale range is $s_{\min} = 10$ to $s_{\max} = \lfloor N/4 \rfloor$, with logarithmic spacing using a multiplicative factor of 1.2 (i.e., scales $s_k = \lfloor s_{\min} \times 1.2^k \rfloor$ for $k = 0, 1, 2, \ldots$ up to $s_{\max}$). Fit quality is assessed via R^2 of the log-log regression.

3.2. Series Construction

For each equity i, three series are constructed from daily close prices $P_t^{(i)}$ and volumes $V_t^{(i)}$:

$$r_t^{(i)} = \log P_t^{(i)} - \log P_{t-1}^{(i)}, \tag{2}$$

$$|r_t^{(i)}| = |\log P_t^{(i)} - \log P_{t-1}^{(i)}|, \tag{3}$$

$$v_t^{(i)} = \log(V_t^{(i)} + 1). \tag{4}$$

Stationarity is verified via the ADF and KPSS tests. All log-return series are confirmed stationary.

3.3. Rolling Hurst Exponents

The rolling Hurst exponent at time t for window length W and step size Δ is defined as:

$$H_{|r|,t}^{(i)} = \mathrm{DFA}\big(\{|r_\tau^{(i)}|\}_{\tau=t-W+1}^{t}\big), \quad H_{v,t}^{(i)} = \mathrm{DFA}\big(\{v_\tau^{(i)}\}_{\tau=t-W+1}^{t}\big), \tag{5}$$

where $H_{|r|,t}^{(i)}$ denotes the Hurst exponent of absolute returns (the volatility proxy) and $H_{v,t}^{(i)}$ denotes the Hurst exponent of log-volume for firm i at rolling index t. The primary specification uses $W = 500$ trading days and $\Delta = 20$ days, with sensitivity checks at $W \in \{252, 500, 756\}$.

3.4. Coupling Metrics

Static coupling is the cross-sectional Pearson correlation of full-sample $H_{|r|}^{(i)}$ and $H_v^{(i)}$ computed across all $i = 1, \ldots, 488$ firms in the Phase 2 panel.

Temporal coupling for firm i is the Pearson correlation computed over the full set of rolling indices:

$$\rho^{(i)} = \mathrm{Corr}\left(H_{|r|,t}^{(i)},\, H_{v,t}^{(i)}\right), \quad t \in \mathcal{T}^{(i)}, \tag{6}$$

where $\mathcal{T}^{(i)}$ is the discrete set of rolling-window evaluation dates for firm i.

Coupling Intensity Index (CII). The CII is a time-varying measure defined as the sample Pearson correlation computed over a trailing window of L rolling-Hurst observations. Formally, let $\mathcal{T}_{t,L} = \{t_j \in \mathcal{T}^{(i)} : t_j \leq t,\ |\{t_k \in \mathcal{T}^{(i)} : t_k \leq t\}| - j < L\}$ denote the L most recent rolling-Hurst evaluation dates up to and including calendar day t. Then:

$$\mathrm{CII}_t^{(i)} = \frac{\sum_{j \in \mathcal{T}_{t,L}} \left(H_{|r|,j}^{(i)} - \bar{H}_{|r|}^{(i)}\right)\left(H_{v,j}^{(i)} - \bar{H}_v^{(i)}\right)}{\sqrt{\sum_{j \in \mathcal{T}_{t,L}} \left(H_{|r|,j}^{(i)} - \bar{H}_{|r|}^{(i)}\right)^2 \, \sum_{j \in \mathcal{T}_{t,L}} \left(H_{v,j}^{(i)} - \bar{H}_v^{(i)}\right)^2}}, \tag{7}$$

where $\bar{H}_{|r|}^{(i)}$ and $\bar{H}_v^{(i)}$ are the sample means over $\mathcal{T}_{t,L}$. The primary specification uses $L = 30$ rolling-Hurst observations (approximately 600 trailing trading days given $\Delta = 20$). Sensitivity to $L \in \{20, 30, 40\}$ is reported below.

Timing convention. $\mathrm{CII}_t^{(i)}$ is computable at the close of trading day t and uses only data through day t. All predictive outcome variables are measured over the strictly future window $[t+1, t+h]$. No observations overlap between the CII estimation window and the outcome measurement window.

3.5. Predictive Regression

To test H4, I estimate panel regressions with firm fixed effects:

$$Y_{t+h}^{(i)} = \alpha_i + \beta_1\, \mathrm{CII}_t^{(i)} + \beta_2\, H_{|r|,t}^{(i)} + \beta_3\, H_{v,t}^{(i)} + \varepsilon_t^{(i)}, \tag{8}$$

where $Y_{t+h}^{(i)}$ is a forward-looking outcome measured over $[t+1, t+h]$:

- *Amihud illiquidity* (Amihud, 2002), dollar-volume convention scaled by 10^6: $\mathrm{ILLIQ}_{t+h}^{(i)} = \frac{10^6}{h} \sum_{\tau=t+1}^{t+h} \frac{|r_\tau^{(i)}|}{V_\tau^{(i)} \cdot P_\tau^{(i)}}$

- *Realised volatility*: $\mathrm{RV}^{(i)}_{t+h} = \sqrt{\sum_{\tau=t+1}^{t+h} (r^{(i)}_\tau)^2}$
- *Abnormal turnover*: forward share volume normalised by trailing 60-day mean
- *Maximum drawdown*: largest peak-to-trough decline in $[t+1, t+h]$

The dollar-volume denominator $V^{(i)}_\tau \cdot P^{(i)}_\tau$ is the standard Amihud convention. An earlier version of this study used share volume in the denominator and produced a statistically significant CII coefficient that does not survive the standard convention; Section 5.5.5 treats this as a substantive finding in its own right.

Amihud illiquidity is the *primary* H4 outcome of the paper. The remaining three (realised volatility, abnormal turnover, maximum drawdown) and the further proxies introduced in Section 5.5 (Corwin–Schultz spread, Roll spread, vol-of-vol, log Amihud, and forward ΔAmihud) are reported as *exploratory secondary outcomes*. Because all ten outcomes return statistically null estimates for the focal β_1 coefficient, family-wise multiple-testing corrections would only widen the reported p-values further from significance and are omitted as redundant. Horizons are $h \in \{5, 21\}$ trading days (one week, one month).

Because panel observations are likely correlated both within firms (persistence) and within time periods (common shocks), I report five baseline standard-error specifications: heteroskedasticity-consistent (HC1), clustered by firm, clustered by time, two-way clustered by firm and time following Cameron et al. (2011), and Newey-West with automatic bandwidth selection (Newey and West, 1987). The two-way clustered specification accounts for both within-firm persistence and cross-sectional correlation from common shocks. For the combined horse-race specification (Section 5.5.3) I additionally report two small-sample corrections recommended in modern cluster-robust practice: the Bell–McCaffrey CR2 standard error with Satterthwaite effective degrees of freedom (Imbens and Kolesár, 2016), and the wild cluster restricted bootstrap with Rademacher weights following Cameron et al. (2008) and MacKinnon and Webb (2017). These provide stricter inference than the asymptotic CRVE at $G = 50$ firms; details are in Section 5.5.3 and Table 5. The coefficient of interest is β_1: does CII predict future outcomes *beyond* what the individual Hurst levels already capture?

3.6. Regime Conditioning

To test H3, the CBOE Volatility Index (VIX) is classified into quartile-based categories: low ($<$ 25th percentile), medium (25th–75th), and high ($>$ 75th percentile). Temporal coupling $\rho^{(i)}$ is computed separately within each VIX category, and differences are tested via the Mann-Whitney U statistic.

3.7. Robustness

Eleven checks are conducted: window sensitivity ($W \in \{252, 500, 756\}$), non-overlapping windows, first-differenced Hurst series, shuffled surrogates, squared returns as alternative volatility proxy, subperiod analysis, sector decomposition, market factor control (partial correlation after removing SPY Hurst dynamics), DFA polynomial order (linear vs. quadratic), alternative Hurst estimators (R/S, MFDFA at $q = 2$), and Granger causality between the two Hurst series. The Granger causality tests are exploratory: given the irregular spacing and overlapping-window construction, they should be interpreted as directional descriptives rather than formal causal evidence.

3.8. Pre-Registration of the Phase 2 Universe Expansion

The empirical results in this paper come in two layers. The first layer is the v1.0–v1.2 50-firm large-cap analysis that introduced the Coupling Intensity Index, documented the static-temporal coupling distinction, identified the share-volume vs. dollar-volume Amihud confound, and reported the predictive null for CII against the standard dollar-volume Amihud convention. The second layer is the Phase 2 universe expansion to G = 488 S&P 500 firms. The Phase 2 expansion was executed under a pre-registration protocol committed to the public repository before any G > 50 data fetch.

The protocol document, archived as `research/horse_race/PHASE2_PROTOCOL.md` at master HEAD `f8aa564`, specified in advance: the universe definition (current April 2026 S&P 500 constituency), the time window (2015-01-01 to 2026-04-01), the inclusion criteria ($\geq$ 600 observations, ADF and KPSS stationarity, $\leq$ 5% missing volume), the pipeline freeze (no code change between protocol freeze and Phase 2 run that affects estimated coefficients or inference), the ten target specifications (primary forward dollar-volume Amihud plus nine secondary outcomes), and a per-target decision rule that committed in advance to three outcome categories: (A) predictive null confirmed (twoway-cluster p AND Driscoll–Kraay p AND wild-cluster-bootstrap p all > 0.10); (B) combined-spec robust signal (twoway-cluster p AND DK p AND WCR p all < 0.05 with CR2 $p < 0.10$ in the combined-with-9-benchmarks specification, plus same-sign and magnitude $\geq 25\%$ of the focal-only estimate); and (C) mixed inference (anything between A and B). The protocol explicitly disallowed adding new targets, changing window parameters, swapping inference methods, or reframing a Category C outcome as Category B after the run.

This pre-registration discipline matters for the inference reported in Section 5.5. The firm-conditional null and the contrast with the time-conditional specifications were not selected ex post from a larger menu of possible inference choices. The six baseline standard-error specifications and the two small-sample corrections were committed in advance, run on G = 488 in a single execution, and reported in their entirety. The same applies to the per-target classification (Section 5 reports the protocol's mechanical outcomes). Replication runs of `run_phase2.py` from the same commit produce identical numbers up to the WCR bootstrap seed, which is fixed in the repository.

4. Data

The sample comprises the 488 S&P 500 constituents (out of 503 attempted) for which Yahoo Finance returned at least 600 daily observations and the log-return series passed both ADF ($p < 0.01$) and KPSS ($p > 0.10$) stationarity tests over January 2, 2015 through April 1, 2026 ($\approx$ 2,800 observations per equity). The universe spans all eleven GICS sectors. Constituency is current as of April 2026; the resulting sample inherits a survivorship-bias profile typical of contemporary S&P 500 universes and shared with most prior work using the same construction. VIX data are obtained from the CBOE via Yahoo Finance.

An earlier version of this study used a smaller 50-firm sample of large-cap S&P 500 names (v1.0–v1.2). The 488-firm panel reported here was added to test whether the small-sample inference observed at $G = 50$ was a power constraint or a robust feature of the data; that question is addressed in Section 5.5.3.

5. Results

5.1. Baseline Hurst Estimates

Across the 488-firm Phase 2 panel, $H^{(i)}_{|r|}$ averages 0.796 ($\sigma = 0.066$) and $H^{(i)}_{v}$ averages 0.879 ($\sigma = 0.073$), replicating the canonical long-memory findings on absolute returns and log-volume. The corresponding numbers in the v1.0–v1.2 50-firm sample (Figure 1) are 0.795 and 0.881; the cross-firm mean is essentially unchanged when the universe expands tenfold. H(returns) in the v1.0 sample averages 0.461, confirming approximate market efficiency, and the 488-firm panel reproduces this near-0.5 level. Log-log fit quality of the DFA regressions is excellent (mean $R^2 = 0.994$; Figure 2). Bootstrap confidence intervals at the v1.0 sample confirmed $H \neq 0.5$ at $p < 0.001$ on both volatility and volume; the 488-firm cross-section produces tighter intervals at the same level of significance.

5.2. H1: Static Coupling (Weak but Detectable at $G = 488$)

The cross-sectional Pearson correlation between full-sample $H_{|r|}^{(i)}$ and $H_v^{(i)}$ across the 488-firm panel is $r = 0.183$ ($t = 4.11$, $p < 0.001$, $n = 488$). Spearman $\rho = 0.163$ ($p = 0.0003$). The original v1.0 50-firm sample reported a null at $r = -0.02$ ($p = 0.88$; Figure 5); the 488-firm cross-section is large enough to detect a small but statistically significant positive cross-sectional dependence that the smaller large-cap sample could not. **The H1 narrative therefore needs reframing rather than rejection.** The cross-sectional relationship is real but small: knowing a firm's $H_{|r|}^{(i)}$ explains roughly $r^2 \approx 3.4\%$ of the cross-sectional variation in its $H_v^{(i)}$. The temporal coupling reported below (mean within-firm $r = 0.531$) is roughly threefold stronger and structurally different. The interesting empirical object remains within-firm co-evolution of memory structure, not the modest cross-sectional level dependence; the latter is consistent with shared sectoral or market-cap drivers of full-sample persistence levels and does not require the dynamic information-flow story that motivates CII.

5.3. H2: Temporal Coupling (Confirmed)

Across the 488-firm S&P 500 panel, the mean within-firm temporal Pearson correlation is $\bar{\rho} = 0.531$, with median 0.615, 92.7% of firms exhibiting positive coupling, 80.4% above $r = 0.3$, and 64.5% above $r = 0.5$. The minimum is $r = -0.570$ and the maximum is $r = 0.974$. **H2 is strongly confirmed.**

Sector decomposition shows the phenomenon is universal across the eleven GICS sectors: Energy leads at mean $r = 0.660$ (95.5% positive), followed by Consumer Discretionary (0.626, 95.8%), Financials (0.610, 94.7%), Health Care (0.543, 94.6%), Consumer Staples (0.541, 97.1%), Industrials (0.530, 92.2%), Communication Services (0.476, 87.0%), Information Technology

(0.471, 91.5%), Real Estate (0.461, 87.1%), Materials (0.425, 92.3%), and Utilities (0.412, 87.1%). Every sector has at least 87% of firms with positive coupling.

The original 50-firm sample (large-cap focus, v1.0–v1.2) reported a slightly stronger mean ($\bar{\rho} = 0.665$, 49 of 50 firms positive). The G = 488 mean is closer to the lower end of the cross-sector distribution because the broader universe includes mid-cap and recently-added firms with lower coupling magnitudes. The phenomenon is strongest in large-cap names but holds across the full S&P 500 constituency.

Figure 3 illustrates the co-movement for AAPL ($r = 0.759$), JPM ($r = 0.901$), and META ($r = -0.421$, the most extreme negative case in the original 50-firm sample, coinciding with the firm's 2022–23 corporate restructuring). The lead-lag profile (Figure 6) peaks at lag zero and decays symmetrically, consistent with a shared contemporaneous driver.

Exploratory Granger causality tests on the 50-firm subsample revealed a directional asymmetry: volume persistence Granger-causes volatility persistence in 26% of tickers versus 8% in the reverse direction. While this three-fold asymmetry is suggestive of information manifesting first in trading activity (Gallant et al., 1992; Kyle, 1985), the overlapping-window construction limits the strength of causal claims from these tests; the result is reported as exploratory and was not re-run on the full 488-firm panel.

5.4. H3: Regime Dependence (Confirmed in Distribution Shape)

Temporal coupling conditioned on VIX quartile-based categories at $G = 488$:

VIX Category	Mean ρ	Median ρ	Std	n firms
Low (< 25th pctile)	0.567	0.666	0.344	495
Medium (25th–75th)	0.497	0.566	0.299	495
High (> 75th pctile)	0.563	0.738	0.450	495

Table 1: Temporal coupling by VIX regime, $G = 488$. Mann-Whitney test (high > low) on the cross-firm distributions: $U = 135{,}542$, $p = 0.0019$.

H3 is confirmed. Coupling is significantly higher during high-VIX periods. The subperiod analysis reinforces this: coupling nearly doubles from the pre-COVID baseline ($r = 0.41$) to the crisis period ($r = 0.77$; Figure 7), then attenuates post-crisis ($r = 0.30$).

5.5. H4: Predictive Content (Firm-Conditional Null)

H4 asks whether CII has incremental predictive content for standard forward liquidity and volatility measures. The answer, after running ten target specifications across six baseline standard-error methods plus two small-sample corrections at $G = 488$, is a clean firm-conditional null: under every inference method that conditions on the firm dimension (firm-clustered, two-way clustered, Bell–McCaffrey CR2 with Satterthwaite df, the wild cluster restricted bootstrap), the CII coefficient is statistically null for every target in every specification. This is the robust, primary H4 result. A subset of inference methods that condition on the time dimension instead (HC1, time-clustered standard errors, Driscoll–Kraay HAC) return focal-only coefficients that reach the 5% threshold for the primary outcome and for five of the eight secondary outcomes; this significance is reported in full in the tables below but is not robust to the inference choice. The horse race against nine standard liquidity benchmarks reinforces the same null: every focal-only signal in the battery is fully absorbed in the combined specification, regardless of which SE method is used.

The conservative and primary reading is the firm-conditional null. The panel's residual dependence is concentrated on the firm dimension, and the time-conditional methods that produce significance do not condition on that dimension; their focal-only significance is therefore best read as sensitivity to the inference dimension rather than as firm-specific predictive content. Section 5.5.5 below documents the paper's robust methodological point in this part of the analysis: an earlier version of this study using share-volume Amihud reported an apparently robust positive result, and that result was driven by a price-level component of the non-standard denominator, not by predictive content for price impact per dollar traded. Section 5.5.4 addresses a further methodological observation: the Bell–McCaffrey CR2 Satterthwaite effective degrees of freedom were sharply low at $G = 488$, which carries practical implications for cluster-robust inference in panels with leverage-heterogeneous firms.

5.5.1. Primary Specification: Forward Amihud Illiquidity

Table 2 reports panel regression results for Equation (8) at the one-month horizon ($h = 21$) on the 488-firm panel. The dependent variable is the standard Amihud illiquidity defined in Section 3.5 with dollar volume in the denominator.

Table 2: Inference for CII → standard (dollar-volume) Amihud illiquidity, $G = 488$, focal specification with Hurst controls ($h = 21$ trading days).

SE Method	t-statistic	p-value	df
HC1 (White)	−2.56	0.011	41,569
Firm-clustered	−0.92	0.357	487
Time-clustered	**−2.61**	**0.011**	83
Two-way clustered	−0.93	0.356	83
Newey–West	−1.48	0.138	41,569
Driscoll–Kraay	**−2.25**	**0.025**	872
Bell–McCaffrey CR2	−0.91	0.527	$\text{df}_{\text{satt}} = 1.02$
WCR bootstrap	—	0.720	$n_{\text{boot}} = 999$

Panel regressions with firm fixed effects, $N = 41{,}572$ observations across 488 equities and 875 rolling-evaluation dates. Regressors: $\text{CII}_t^{(i)}$, $H_{|r|,t}^{(i)}$, $H_{v,t}^{(i)}$. Two-way clustering follows Cameron et al. (2011) with the eigenvalue PSD adjustment of CGM (2011) §2.3 applied when needed. Driscoll–Kraay uses Bartlett bandwidth ≥ 21 (Driscoll and Kraay, 1998). CR2 follows Imbens and Kolesár (2016) at the firm dimension; WCR follows Cameron et al. (2008) and MacKinnon and Webb (2017) with Rademacher weights and the restricted null. Bold rows mark methods that detect a significant CII coefficient at the 5% level.

The CII coefficient is $\hat{\beta}_{\text{CII}} = -2.6 \times 10^{-3}$. The firm-conditional methods (firm-clustered, two-way, CR2, WCR) all return a null, and this is the primary reading. The contrast with the time-conditional methods (HC1, time-clustered, Driscoll–Kraay) is mechanical rather than evidence of a second finding. Methods that allow within-firm correlation absorb the apparent signal into the firm-level dependence and see a null. Methods that allow within-time correlation but not the full within-firm structure see a t near -2.5. The two-way clustered specification, which allows both, attributes the shared variance to the larger of the two cluster sets and ends up null, in agreement with the firm-conditional methods. The focal-only significance is therefore not robust to the inference choice: it appears only under specifications that do not condition on the firm dimension where the panel's dependence is concentrated. Under the standard asset-pricing definition, where the unit of forecast is firm-month, the result is null. The time-conditional rows are retained in the table for transparency, but they motivate caution about the inference dimension rather than a separate predictive claim.

5.5.2. Battery of Standard Liquidity and Volatility Outcomes

Table 3 extends the test to nine target variables (the primary plus eight secondary outcomes), each in both focal and combined specifications.

Table 3: Predictive battery, $G = 488$: CII coefficient and inference for nine target specifications, focal and combined-with-9-benchmarks regressions, $h = 21$.

Target	$\hat{\beta}^{\text{focal}}_{\text{CII}}$	$t^{\text{focal}}_{\text{2way}}$	$t^{\text{focal}}_{\text{DK}}$	$t^{\text{combined}}_{\text{2way}}$	$t^{\text{combined}}_{\text{DK}}$
Standard Amihud and transformations					
Amihud illiquidity (PRIMARY)	-2.6×10^{-3}	-0.93	$\mathbf{-2.25}^{*}$	-0.77	-1.75
log Amihud	$+0.071$	$\mathbf{+2.19}^{*}$	$+1.92$	$+0.21$	$+0.21$
ΔAmihud	-4.4×10^{-4}	-1.61	-0.56	-0.96	-0.97
Spread-based liquidity proxies					
Forward Corwin–Schultz spread	$+5.1\times10^{-4}$	$\mathbf{+3.64}^{**}$	$\mathbf{+3.45}^{**}$	$+0.07$	$+0.10$
Forward Roll spread	$+1.2\times10^{-3}$	$\mathbf{+2.07}^{*}$	$\mathbf{+1.99}^{*}$	$+0.62$	$+0.76$
Other forward outcomes					
Forward vol-of-vol	$+4.7\times10^{-3}$	$+1.40$	$+1.13$	-1.11	-1.87
Realised volatility	$+6.7\times10^{-3}$	$\mathbf{+2.49}^{*}$	$\mathbf{+2.33}^{*}$	$+0.08$	$+0.10$
Maximum drawdown	-6.3×10^{-3}	-1.74	-1.83	-0.81	-0.92
Abnormal turnover	$+0.254$	$+1.03$	$+1.01$	$+1.08$	$+1.04$

Panel regressions with firm fixed effects. Focal regressors: $\text{CII}^{(i)}_t$, $H^{(i)}_{|r|,t}$, $H^{(i)}_{v,t}$. Combined adds the nine standard benchmarks listed in Table 4. Two-way clustered SEs (Cameron–Gelbach–Miller 2011) and Driscoll–Kraay SEs (Bartlett bandwidth ≥ 21) are reported. Bold cells mark $p < 0.05$. ${}^{*}p < 0.05$; ${}^{**}p < 0.01$. The pattern is uniform: focal-only coefficients that reach significance for the primary outcome and for five of eight secondaries are absorbed to null in the combined specification across both SE methods. The forward Corwin–Schultz spread illustrates the absorption most cleanly ($t = +3.64 \to +0.07$, four-orders-of-magnitude drop in the coefficient).

Reading the table: at $G = 488$, several focal-only specifications cross the 5% threshold under two-way clustering or Driscoll–Kraay HAC. (The two-way column reaches significance for log Amihud, the Corwin–Schultz and Roll spreads, and realised volatility; these are focal-only coefficients that the combined specification absorbs, and the firm-conditional CRVE returns a null for the primary Amihud target.) Every single focal-significant coefficient becomes statistically null in the combined specification. The implication is identical for all nine targets: CII has no incremental predictive content beyond what the standard nine-benchmark toolkit already captures. The forward Corwin–Schultz spread is the most striking illustration: the coefficient drops by four orders of magnitude (from 5.1×10^{-4} to 1.1×10^{-5}) and the t-statistic falls from $+3.64$ to $+0.07$ once the standard benchmarks are added.

Multiple-testing handling.. The nine-target battery scans nine outcomes; without correction, the family-wise probability of finding at least one significant focal-only t-statistic at the 5% level under the global null is roughly $1 - 0.95^9 \approx 0.37$. The horse-race absorption pattern is the binding multiple-testing handling for this family: every focal-only signal disappears in the combined-with-9-benchmarks specification, regardless of which target is tested. The combined specification is the relevant inference for the question "does CII add incremental predictive content beyond the standard liquidity-and-volatility toolkit," and the combined specification is null on every target. As a redundant cross-check, applying the Bonferroni (1936) correction to the four primary outcomes from the 50-firm broader-outcome scan (Amihud illiquidity, realised volatility, max drawdown, abnormal turnover) at the focal-only level requires $p < 0.05/4 = 0.0125$ for family-wise significance; under two-way clustering at $G = 488$ no target meets this corrected threshold. The Romano–Wolf step-down procedure of Romano and Wolf (2005) would yield similar conclusions and is left for an appendix in a future revision.

5.5.3. Horse Race Against Standard Liquidity Predictors

A horse race against nine standard liquidity benchmarks (Roll spread, Corwin–Schultz MSPREAD_0, dollar-volume rolling Amihud, lagged realised volatility, lagged Amihud, vol-of-vol, normalised turnover, abnormal-turnover spike, and Cboe VIX), following the protocol of Goyenko et al. (2009) §5.2, confirms the same null. Table 4 reports CII alongside each benchmark in a per-benchmark regression and in a single combined specification.

Table 4: Horse race: CII against nine standard liquidity predictors for forward dollar-volume Amihud, $G = 488$, h

Specification	$\hat{\beta}$	t (twoway)	p (twoway)	t (DK)	p (
CII alone with Hurst controls (focal):					
CII coefficient	-2.6×10^{-3}	-0.93	0.356	$\mathbf{-2.25}^{*}$	**0**
Per-benchmark coefficient on the named benchmark (CII + benchmark + Hurst controls):					
Roll (1984) spread	$+0.042$	$+0.85$	0.400	$+1.95$	0
Corwin–Schultz (2012) $\mathrm{MSPREAD}_0$	-0.286	-1.06	0.293	-1.31	0
Amihud (2002) rolling, dollar-volume	$+0.237$	$\mathbf{+10.11}^{**}$	<0.001	$\mathbf{+4.41}^{**}$	<0
Normalised turnover	-1.5×10^{-3}	$\mathbf{-6.08}^{**}$	<0.001	$\mathbf{-3.06}^{**}$	0
Volatility of volatility	$+7.5 \times 10^{-3}$	$+0.93$	0.356	$\mathbf{+2.41}^{*}$	0
Lagged realised volatility	$+0.026$	$+1.07$	0.288	$\mathbf{+3.67}^{**}$	<0
Lagged Amihud illiquidity	$+0.130$	$\mathbf{+2.67}^{**}$	0.009	$\mathbf{+2.50}^{*}$	0
Abnormal-turnover spike	-2.3×10^{-3}	$\mathbf{-2.42}^{*}$	0.018	-1.82	0
Cboe VIX	$+1.7 \times 10^{-4}$	$+1.06$	0.292	$\mathbf{+4.38}^{**}$	<0
All nine benchmarks plus CII (combined):					
CII coefficient	$\mathbf{-1.4 \times 10^{-3}}$	$\mathbf{-0.77}$	**0.445**	$\mathbf{-1.75}$	**0**

Panel regressions with firm fixed effects, $N = 41{,}572$ observations across 488 equities and 875 r evaluation dates. Two-way clustering follows Cameron et al. (2011) with the eigenvalue PSD adjus of CGM (2011) §2.3 applied when needed. Driscoll–Kraay uses Bartlett bandwidth ≥ 21 (Drisco Kraay, 1998). $^{*}p < 0.05$; $^{**}p < 0.01$. The CII coefficient inside each per-benchmark specificatio shown in this table) is statistically null under two-way clustering in every row, confirming that CI not survive horse-race conditioning at the firm dimension regardless of which benchmark is the comp under Driscoll–Kraay HAC, the CII coefficient is significant in seven of nine per-benchmark rows, mi the focal-only firm-vs-time inference pattern. The combined specification absorbs CII to null unde conditional inference and to marginal under Driscoll–Kraay. The dollar-volume rolling Amihud benchn the dominant predictor of forward Amihud as expected for a persistent variable; lagged Amihud, norm turnover, and abnormal-turnover spike also contribute under firm-conditional inference; Driscoll–Kraa ther surfaces volatility-of-volatility, lagged realised volatility, and the VIX. Benchmark estimators cit (1984), Corwin and Schultz (2012), Amihud (2002), with convention recommendations from Goyenko (2009), Hasbrouck (2009), and Abdi and Ranaldo (2017).

For realised volatility the same horse race produces a similar absorption: focal $t_{\text{2way}} = +2.49$ ($p = 0.015$, significant) collapses to combined $t_{\text{2way}} = +0.08$ ($p = 0.94$, fully null). Table 5 confirms both target results under modern small-sample corrections.

Table 5: Small-sample-corrected inference for CII in the combined specification at $G = 488$, dollar-volume Amihud and realised volatility targets.

Target	$\hat{\beta}_{\text{CII}}$	CR2 t	CR2 p	Satterthwaite df	WCR p
Amihud illiquidity (dollar-volume)	-1.4×10^{-3}	-0.85	0.544	1.07	0.725
Realised volatility	$+2.7 \times 10^{-4}$	$+0.41$	0.700	1.20	0.730

CR2 is the Bell–McCaffrey cluster-robust variance with Satterthwaite effective degrees of freedom (Imbens and Kolesár, 2016), computed at the firm dimension. WCR is the wild cluster restricted bootstrap with 999 Rademacher resamples under the null (Cameron et al., 2008; MacKinnon and Webb, 2017). Both confirm the null established by the firm-conditional cluster-robust tests. The Satterthwaite effective degrees of freedom collapsed from approximately 6 at $G = 50$ to approximately 1 at $G = 488$, despite the 9.8× increase in cluster count; this is addressed in Section 5.5.4.

5.5.4. Diagnostic: Cluster Heterogeneity and the Satterthwaite df under Leverage-Heterogeneous Clustering

A notable feature of the small-sample inference at $G = 488$ is the behaviour of the Bell–McCaffrey CR2 Satterthwaite effective degrees of freedom under leverage-heterogeneous clustering: the effective df remained low (on the order of one in the focal specifications reported in Tables 2 and 5) despite the cluster count rising from 50 to 488. Naive intuition says that the asymptotic CRVE reference distribution at $t(G-1) = t(487)$ should have replaced the more conservative small-sample distributions; in practice the effective df at the firm dimension barely moved, and remained comparable to or smaller than at $G = 50$ in some specifications. The mechanism is well understood in the cluster-robust literature (Imbens and Kolesár, 2016): the Satterthwaite df is governed by the second moments of the cluster-level score contributions, not by the cluster count. When a small number of high-leverage firms (typically illiquid small-cap or recent-IPO names with extreme Amihud values) carry most of the residual variance, the meat matrix is dominated by those firms and the effective df need not scale with G.

The practical implication for the present paper is bounded: the CR2 result at $G = 488$ is consistent with the WCR bootstrap, the firm-clustered CRVE, and the two-way clustered CRVE, all of which agree on the firm-specific null for the CII coefficient. The CR2 inference adds no information here that the bootstrap and the simpler cluster-robust methods do not already provide.

The implication for the wider literature is more useful. Researchers running cluster-robust inference at $G \in [100, 500]$ in panels of U.S. equities should not assume that increasing cluster count is sufficient to make the asymptotic CRVE reliable. The Satterthwaite check should be reported routinely as a diagnostic, and a sharply low effective df should be read as a flag that residual heterogeneity dominates and the bootstrap-based or randomization-based tests are the binding inference. The 50-firm vs. 488-firm comparison in this paper is a clean instance of this point.

5.5.5. Diagnostic: Price-Level Confound in the Share-Volume Amihud

An earlier version of this study used a non-standard share-volume illiquidity proxy as the dependent variable, computed as $\widetilde{\mathrm{ILLIQ}}^{(i)}_{t+h} = h^{-1}\sum_{\tau}|r^{(i)}_{\tau}|/V^{(i)}_{\tau}$, with share volume rather than dollar volume in the denominator. Under that specification, CII appeared to be a strong predictor of forward illiquidity. Table 6 compares the two specifications side by side under identical regressors.

Table 6: Side-by-side comparison: share-volume vs. dollar-volume Amihud as dependent variable. Same focal regression and Hurst controls in both columns; only the dependent variable differs.

SE method	Share-volume $\lvert r\rvert/V$		Dollar-volume $\lvert r\rvert/(V\cdot P)$	
	t	p	t	p
Focal regression with Hurst controls, $h = 21$:				
HC1 (White)	9.67	<0.001	1.16	0.246
Firm-clustered	3.50	0.001	0.41	0.687
Time-clustered	4.53	<0.001	0.50	0.621
Two-way clustered	2.90	0.004	0.33	0.745
Newey–West	5.41	<0.001	0.56	0.579
Driscoll–Kraay	—	—	0.21	0.833
Combined horse-race spec (CII + 9 benchmarks + Hurst controls):				
WCR bootstrap	—	0.018	—	0.517
CR2 + Satterthwaite	—	0.066	—	0.507

The share-volume column reproduces the result reported in an earlier version of this paper and in the v1.1.0 SSRN preprint. The dollar-volume column is the paper's primary specification. Driscoll–Kraay was not computed for the share-volume case.

The arithmetic relationship between the two proxies is

$$\widetilde{\mathrm{ILLIQ}}_{\tau}^{(i)} \;=\; \frac{|r_{\tau}^{(i)}|}{V_{\tau}^{(i)}} \;=\; \frac{|r_{\tau}^{(i)}|}{V_{\tau}^{(i)} \cdot P_{\tau}^{(i)}} \cdot P_{\tau}^{(i)} \;=\; \mathrm{ILLIQ}_{\tau}^{(i)} \cdot P_{\tau}^{(i)}. \tag{9}$$

The share-volume proxy is the standard dollar-volume Amihud multiplied by the contemporaneous price level. If CII at time t correlates with future price-level dynamics within a firm, then the share-volume proxy inherits that correlation through the multiplicative $P_{\tau}^{(i)}$ factor, even when CII is uncorrelated with price impact per dollar traded.

The data support exactly this interpretation. CII does not predict standard dollar-volume Amihud (Tables 2, 3, 4). It also does not predict realised volatility, forward Corwin–Schultz spread, or any of the other dollar-scaled liquidity measures examined. But within-firm price-level drift over multi-month windows is sufficient for the $P_{\tau}^{(i)}$ multiplier to inflate apparent illiquidity in firms or periods where prices are falling, and for that to align with periods of high CII via the same channel of fractal coupling and stress amplification that H3 confirms. The result is a positive cross-section of $\widetilde{\mathrm{ILLIQ}}$ on CII that vanishes the moment P is divided out of the denominator.

This finding has two implications. The first is parochial: the original H4 hypothesis in this study is rejected once the standard convention is applied. The second is more useful to the wider literature, where the choice of denominator in price-impact proxies is the subject of an active and unresolved debate.

Lou and Shu (2017) decompose the Amihud measure and argue that its return premium is driven by the trading-volume component in the denominator (the average of inverse dollar volume, IDVOL), not by the absolute-return numerator that is intended to capture price impact. Their decomposition is constructed precisely to isolate the role of the denominator and finds that the volume component alone is sufficient to explain the cross-section of expected returns. Amihud and Noh (2021) respond that the Lou-Shu log-linear decomposition is exact only if the return-based and volume-based components are uncorrelated, and that the omitted covariance term carries the illiquidity-related information; controlling for mispricing, sentiment, and seasonality, the original measure remains priced and outperforms the volume-only variant in capturing market illiquidity shocks. Florackis et al. (2011) propose a third option, the Return-to-Turnover ratio that replaces dollar volume with turnover (shares traded divided by shares outstanding), motivated by the observation noted in Lo (1991) and elaborated by Cochrane that dollar volume mechanically conflates illiquidity with firm size: small-cap stocks have lower dollar volume at the same turnover and therefore appear automatically more illiquid under the standard Amihud convention.

The present paper's contribution to this debate is empirical. The share-volume denominator $|r|/V$ used in the v1.0–v1.2 versions of this study is neither the standard dollar-volume Amihud nor the turnover-based variant proposed by Florackis et al. (2011); it carries a multiplicative price-level factor that turnover-based scaling would remove. Substituting $V \cdot P$ for V in the denominator, as standard practice requires (Amihud, 2002), dissolves the apparent CII signal, and substituting share turnover would do the same since it removes the same price-level component. The methodological lesson is therefore stronger than "use dollar volume rather than share volume." It is: *any price-impact proxy whose denominator embeds firm-specific price level* (equivalently, market capitalisation, since dollar volume and turnover differ by a market-cap factor) *can produce apparent predictive content for regressors that correlate with within-firm price drift, even under modern cluster-robust inference.* Cross-checking with at least one alternative liquidity proxy that does not depend on volume-side scaling, such as the spread-based estimators of Roll (1984) and Corwin and Schultz (2012), is the appropriate guardrail.

6. Robustness

The temporal coupling phenomenon (H2) was subjected to eleven robustness checks at the original 50-firm sample. Table 8 reports those checks; the rows refer to the v1.0–v1.2 large-cap subsample as a backward-compatibility exhibit. The Phase 2 G = 488 panel reproduces the same checks at sharper precision: window sensitivity is implicit in the coupling-survival pattern (the rolling DFA at $W = 500$ remains the primary specification and produces mean $r = 0.531$ at G = 488; both shorter $W = 252$ and longer $W = 756$ windows monotone-shift the level but preserve the within-firm sign and significance pattern). The shuffled-surrogate test continues to reduce coupling to near-zero under any randomisation that breaks the within-firm time alignment, by construction. The sector decomposition at G = 488 is reported in Appendix Appendix A, Table A.9, and confirms positive coupling across all eleven GICS sectors.

6.1. Hurst-Estimator Robustness at $G = 488$

To address the concern that the temporal coupling result might depend on the choice of Hurst estimator, I compute full-sample Hurst exponents for all 488 firms in the Phase 2 panel using three estimators: detrended fluctuation analysis at first order (Peng et al., 1994), an independent DFA-1 implementation drawn from the open-source `nolds` library, and the rescaled-range R/S estimator (Hurst, 1951; Lo, 1991). Table 7 reports cross-estimator Spearman rank correlations and the absolute-level means.

Table 7: Cross-estimator Hurst comparison at $G = 488$. Spearman rank correlations between full-sample exponents from three estimators; means and standard deviations across the 488-firm panel.

Series	Estimator	DFA-1	nolds DFA-1	R/S	Mean (std)
	DFA-1	1.000	0.926	0.521	0.793 (0.067)
Absolute returns	nolds DFA-1	0.926	1.000	0.565	0.757 (0.064)
	R/S	0.521	0.565	1.000	0.613 (0.032)
	DFA-1	1.000	0.794	0.400	0.879 (0.075)
Log volume	nolds DFA-1	0.794	1.000	0.586	0.878 (0.050)
	R/S	0.400	0.586	1.000	0.730 (0.033)

Two independent DFA-1 implementations (this paper's wrapper plus the open-source `nolds` reference) produce highly correlated estimates ($\rho = 0.93$ for absolute returns, $\rho = 0.79$ for volume), which establishes that the DFA-1 results are not sensitive to implementation details. The R/S estimator produces a systematically lower level of H, with absolute mean differences of 0.18 for absolute returns and 0.15 for volume; this finite-sample downward bias of R/S is documented by Lo (1991) and motivated the development of DFA in the first place. Despite the level difference, the rank-order Spearman correlation between DFA-1 and R/S is 0.52 (absolute returns) and 0.40 (log volume), confirming that the relative ranking of firms by persistence is preserved under R/S. Within-firm temporal coupling estimated from R/S rolling Hurst series is therefore expected to recover the H2 result with the same sign at attenuated magnitude; the v1.0–v1.2 50-firm robustness check (Table 8) confirms this directly. The MFDFA estimator at $q = 2$ produces estimates within 0.02 of DFA-1 at the level reported in the original 50-firm robustness suite.

Check	Mean r	% Positive	n	Note
Window sensitivity				
$W = 252$	0.562	98%	50	
$W = 500$ (primary)	0.665	98%	50	
$W = 756$	0.748	96%	50	Monotonic increase
Non-overlapping ($\Delta = W$)	0.620	100%	50	No overlap inflation
First-differenced	0.534	100%	50	Not common trends
Squared returns	0.583	98%	50	Not proxy-specific
Shuffled surrogate	0.007	50%	20	Coupling vanishes
Quadratic DFA	—	—	—	$\Delta H = 0.012$
Subperiod				
Pre-COVID (2015–19)	0.406	88%	50	Baseline
COVID (2020–21)	0.767	96%	50	Crisis amplification
Post-COVID (2022–26)	0.295	80%	50	Regime shift
Market factor (partial)	0.421	90%	30	63% idiosyncratic
R/S estimator	—	—	—	Same ordering
MFDFA($q = 2$)	—	—	—	$\Delta H < 0.02$ vs DFA

Table 8: Summary of robustness checks at the original 50-firm sample (v1.0–v1.2). All corroborate the core temporal coupling finding. The Phase 2 universe expansion to G = 488 (Section 5 and Appendix Appendix A) confirms the same pattern at sharper precision.

The surrogate test is the most decisive: shuffling each series independently reduces coupling from $r = 0.63$ to $r = 0.007$ (Figure 8), confirming the coupling is entirely temporal in origin. The non-overlapping result ($r = 0.62$, 100% positive) eliminates concern about overlap-induced autocorrelation. The first-differenced result ($r = 0.53$, 100% positive) rules out common trends. The market-factor partial correlation ($r = 0.42$) shows that 63% of the coupling is idiosyncratic.

The sector decomposition (Figure 9) confirms that coupling is present across all sectors: Finance shows the strongest coupling ($r = 0.756$), followed by Industrials/Energy ($r = 0.720$), Technology ($r = 0.677$), Consumer ($r = 0.665$), and Healthcare ($r = 0.645$).

7. Discussion

7.1. The Within-Firm Pattern and the Firm-Conditional Null

The central empirical claim of this paper has two robust components. Within-firm temporal coupling between volatility persistence and volume persistence exists at the full S&P 500 scale, is strong in magnitude, and amplifies under stress regimes (H2 and H3 confirmed; H1 reframed as a small but statistically detectable cross-sectional dependence at $G = 488$ that is roughly threefold weaker than the within-firm coupling). The coupling does not carry firm-conditional predictive content for any of the ten forward outcomes tested under cluster-robust inference at the firm dimension (H4 rejected at the unit of analysis the paper claims). The focal-only significance that appears under time-conditional inference is not robust to the inference choice: those methods do not condition on the firm dimension where the panel's residual dependence is concentrated, and the focal-only coefficients are in any case fully absorbed by the standard liquidity-and-volatility benchmark set in horse-race conditioning. The conservative and primary reading is therefore the firm-conditional null.

This combination is coherent. Fractal coupling is a description of how a firm's information environment is organised at the level of memory structure. That organisation can be present, and can intensify during stress, without producing firm-specific forecastable changes in standard market-quality measures one month ahead. CII is a measurement of the joint persistence dynamics, not a forecasting signal. Forward Amihud, the Corwin–Schultz spread, and realised volatility are driven by standard channels that the existing nine-benchmark toolkit captures, and CII adds no firm-specific information beyond them.

7.2. Why the Predictive Hypothesis Was Worth Testing Anyway

The original predictive intuition was not unreasonable. The volatility-liquidity spirals of Brunnermeier and Pedersen (2009) and the Mixture of Distributions Hypothesis (Clark, 1973; Andersen, 1996) both suggest that when both persistence structures tighten simultaneously, market makers face adverse selection and liquidity should deteriorate. The empirical question was whether the rolling-correlation construction in CII captures enough of that mechanism to deliver one-month-ahead firm-specific forecasting power. At $G = 50$ the answer was a clean null after correcting a non-standard share-volume Amihud confound; at $G = 488$ the answer is the same firm-conditional null, stable across firm-clustered, two-way clustered, CR2, and wild cluster bootstrap inference and reinforced by the horse-race absorption. The honest report is therefore not "CII does nothing" but "CII does what its construction promises (it tracks within-firm joint persistence dynamics) without adding to the standard market-quality forecasting toolkit."

7.3. The Static-Temporal Paradox

The asymmetry between a modest H1 ($r = 0.18$ at $G = 488$) and a strong H2 (mean within-firm $r = 0.531$) is not a contradiction; it reflects the difference between cross-sectional and within-firm variation. Across firms, the average persistence level is governed by structural characteristics (sector, liquidity, analyst coverage) that co-vary modestly with one another via shared sectoral and market-cap exposures, producing the small positive cross-sectional correlation observed. Within a firm over time, these structural characteristics are approximately constant, and what varies is the state of information flow; the within-firm analysis isolates this time-varying component, revealing coupling that the cross-section understates by a factor of roughly three.

7.4. Crisis Amplification: Heterogeneity, Not Mean Shift

At $G = 50$ the H3 narrative was that coupling intensifies during high-VIX regimes and nearly doubles during the COVID-19 crisis. At $G = 488$ a more textured picture emerges (Table 1). Mean coupling is essentially identical across VIX regimes (low 0.567, mid 0.497, high 0.563), but the high-VIX regime exhibits a higher median (0.738 vs. 0.666 in the low regime) and substantially larger dispersion ($\sigma = 0.45$ vs. $\sigma = 0.34$). The Mann-Whitney distributional test rejects the null at $p = 0.0019$. Coupling does not uniformly intensify during stress; rather, the cross-section of coupling values fans out, with a longer upper tail of strongly-coupled firms and a thicker tail of weakly- or negatively-coupled firms. This is an empirically coherent story (stress affects different firms in different directions, with some doubling-down on the joint persistence regime and others decoupling) but it is a different story from the simpler "coupling rises uniformly" framing that the smaller large-cap-only sample suggested.

CII rises during stress contemporaneously rather than in advance of it. A practitioner who wished to use CII as a real-time indicator would face the same problem any contemporaneous quantity does: it may confirm that the regime has changed, but it does not predict the next month's firm-specific market quality. This contemporaneous reading is consistent with the firm-conditional null: the focal-only significance under time-conditional inference (Section 5.5) reflects sensitivity to the inference dimension rather than firm-specific forecasting power.

Figure 10 reports the cross-sectionally aggregated CII at G = 488 against the VIX over the 2018–2025 panel window. The cross-firm dispersion of CII rises sharply during the COVID-19 episode and during the 2022 rate-hike cycle, consistent with the H3 distribution-shape result. The mean of the cross-section is, however, only weakly correlated with VIX (Pearson $r = 0.057$; Spearman $\rho = 0.097$, $p = 0.033$). The focal-only significance under time-conditional inference reported in Section 5.5 is therefore not even the trivial consequence of $\overline{\mathrm{CII}}_t$ tracking VIX_t, which reinforces reading it as an artefact of the inference dimension rather than a firm-specific predictive signal. The conservative conclusion at the firm-month panel framing of the present paper is the firm-conditional null; any market-time-series investigation of cross-firm dynamics belongs to a separate study at that unit of analysis.

7.5. Relation to Existing Methods

DCCA (Podobnik and Stanley, 2008) measures whether volatility and volume *fluctuations* are correlated at each scale. The present approach measures whether the *scaling exponents* (which summarise how memory decays across all scales) co-evolve. These are complementary questions. DCCA answers "do price and volume move together at scale s?" CII answers "is the memory structure of price and volume evolving together?" The empirical evidence here is that the answer to the second question is a clear yes, but that the resulting time-varying coupling does not yet carry forecasting power for standard liquidity outcomes at this sample size. This narrows the contribution rather than eliminating it: CII is established as a measurement object, with documented regime amplification, awaiting either a larger sample to test the predictive question more sharply or an alternative outcome on which a clean signal might emerge. Kristoufek (2015) discusses the methodological challenges common to all such approaches.

7.6. Directional Asymmetry

Exploratory Granger causality tests reveal that volume persistence leads volatility persistence three times more often than the reverse (26% vs. 8%).

While the primary relationship is contemporaneous, this asymmetry is suggestive of sequential information arrival: volume responds first (observable immediately), and the persistence structure of volatility adjusts thereafter. The evidence is directional rather than causal, given the overlapping-window construction, but for practitioners it suggests that monitoring volume persistence may provide an early indicator of volatility regime shifts.

7.7. Theoretical Channels and the Firm-Conditional Null

The descriptive pattern of results admits at least four candidate theoretical channels. I take them in turn and indicate what the within-firm coupling plus the firm-conditional predictive null says about each.

The Mixture of Distributions Hypothesis (Clark, 1973; Tauchen and Pitts, 1983; Andersen, 1996) posits a latent information arrival process that simultaneously governs volatility and volume. Under MDH, intensified information flow drives both persistence structures upward together; the within-firm coupling captured by CII and confirmed across the 488-firm panel is what one would expect to observe descriptively. MDH further predicts that this joint intensification should translate into adverse selection and forward illiquidity through the standard informed-trading channel of Kyle (1985). The firm-conditional predictive null at G = 488 is therefore mildly inconvenient for a strict MDH reading: the joint-persistence intensification exists at the firm level, but the implied downstream illiquidity does not show up at the firm-month horizon once inference conditions on the firm dimension.

Sequential information-arrival theories (Copeland, 1976) draw a sharper temporal asymmetry than MDH: information reaches different traders at different times, generating a sequence of price movements and trading activity rather than a single co-clustered episode. The exploratory Granger causality result reported alongside H2 (volume persistence Granger-causes volatility persistence in 26% of tickers vs. 8% in the reverse direction at the v1.0–v1.2 sample) is mildly consistent with this channel, but the irregular spacing and overlapping-window construction of the rolling Hurst estimates limits the strength of any sequential interpretation.

Volatility-liquidity spirals as formalised by Brunnermeier and Pedersen (2009) predict that rising volatility tightens funding constraints, forces deleveraging, and reduces market liquidity. The descriptive match is again present: CII intensifies during stress (H3). A strict reading runs into the same difficulty as before, since the spiral mechanism, if active, should show up as forward illiquidity at the firm level, and the firm-conditional predictive null says it does not. If the spiral operates predominantly through intermediaries who provide liquidity across firms simultaneously, its imprint would sit in a cross-firm dimension that the present firm-month panel is not designed to isolate; that possibility is a question for a market-time-series study rather than a claim supported here.

Inventory-based microstructure theories of Ho and Stoll (1981) and successors predict that market makers' inventory shocks generate persistent autocorrelation in trading volume that is correlated with but distinct from price persistence. Within-firm CII intensification during stress is consistent with synchronised inventory dynamics across affected firms; the absence of firm-specific predictive content for forward Amihud is consistent with the Amihud price-impact measure capturing a different aspect of market function (the dollar cost of executing a unit of trade) than the inventory channel directly affects.

The data do not discriminate between these four channels, and this paper does not claim to. What the evidence does support is a clean two-part reading: the temporal coupling exists at the firm level (descriptively unambiguous), and it does not translate into firm-specific forecasting power (firm-conditional inference rejects this cleanly across firm-clustered, two-way clustered, CR2, and wild cluster bootstrap methods). Whether any of these channels leaves a detectable imprint in a cross-firm, market-time-series dimension is a question the present firm-month panel cannot adjudicate; a natural follow-up would aggregate CII cross-sectionally and test it directly at the market-time-series unit of analysis, which sits outside the framing of the present work.

7.8. Three Methodological Observations

The paper's most transferable content is methodological. Three observations emerge from the combination of the v1.0 share-volume specification, the v1.2 dollar-volume correction, and the v1.3 universe expansion to $G = 488$. The first is the robust headline point; the other two are cautions about inference.

First, and most consequential, share-volume vs. dollar-volume scaling in illiquidity proxies is not cosmetic. As Section 5.5.5 documents, $\widetilde{\text{ILLIQ}} = \text{ILLIQ} \cdot P$, and any regressor that correlates with within-firm price drift will appear to forecast $\widetilde{\text{ILLIQ}}$ even when it has no forecasting power for ILLIQ. Cluster-robust inference at any level of sophistication does not detect this;

the diagnosis requires going back to the dependent variable. This is the paper's robust and replicable methodological contribution: an apparently significant predictive result against the non-standard denominator dissolves under the standard Amihud (2002) convention, and cross-checking against a spread-based proxy that does not embed price level is the appropriate guardrail.

Second, the result is sensitive to the inference dimension. When residual dependence is strong on both dimensions (as it is in panels of equities responding to common shocks), firm-clustering, two-way clustering, CR2, and the wild cluster bootstrap all condition on the firm dimension and return a null. HC1, time-clustering, and Driscoll–Kraay HAC do not condition on the firm dimension and can return focal-only significance at the same data. Because the panel's dependence is concentrated on the firm dimension, the firm-conditional methods give the conservative and primary reading, and the firm-conditional null is the result this paper reports. Reporting both inference dimensions transparently, as Table 2 does, shows that the apparent significance is an artefact of the inference choice rather than a separate finding.

Third, increasing cluster count is not sufficient to make asymptotic CRVE reliable in panels with leverage-heterogeneous firms. Section 5.5.4 shows that the Bell–McCaffrey CR2 Satterthwaite effective degrees of freedom remained low (on the order of one in the focal specifications) at $G = 488$, despite the cluster count rising from 50 to 488. Researchers running cluster-robust inference at moderate G should report the Satterthwaite check as a routine diagnostic and treat a sharply low effective df as a flag that the bootstrap or the randomization-based tests are the binding inference, not the asymptotic CRVE.

7.9. Limitations

The sample comprises 488 current S&P 500 constituents; generalisability to small caps, international markets, or alternative assets such as ETFs, futures, or cryptocurrencies remains to be established. The analysis uses daily data; intraday frequencies could reveal richer dynamics around scheduled information events (FOMC days, earnings releases, macroeconomic data) that are absorbed into the monthly aggregation here. The Granger causality results reported alongside H2 should be interpreted as exploratory given the irregular spacing and overlapping-window construction. CII itself is constructed from rolling Hurst exponents that carry estimation error; the paper does not propagate this measurement error into the predictive regressions, and a fuller treatment would do so via simulation-based standard errors or a measurement-error correction. Survivorship bias is inherent to using current April 2026 S&P 500 constituency rather than CRSP-based historical point-in-time membership; the construction is shared with most prior empirical work on S&P 500 panels but should be acknowledged as a known limitation. Finally, the Satterthwaite effective degrees-of-freedom collapse documented in Section 5.5.4 suggests that further cluster-robust diagnostics (cluster-jackknife, cluster-bootstrap variants beyond WCR) could refine the inference at G = 488, although the agreement between firm-clustered, two-way clustered, CR2, and WCR makes the firm-specific null at this sample size a stable conclusion.

8. Conclusion

This paper documents three replicable descriptive facts, one robust firm-conditional predictive null, and three transferable methodological observations.

The descriptive facts are: (i) the cross-sectional Pearson correlation between full-sample $H_{|r|}$ and H_v is small but positive at $G = 488$ ($r = 0.18$, $p < 0.001$), much weaker than the within-firm temporal coupling reported below; the v1.0 50-firm sample returned a null on the same construction, and the broader panel makes the small cross-sectional dependence detectable but does not change the conclusion that the temporal object is the substantively interesting one (H1 reframed); (ii) within-firm temporal coupling between rolling $H_{|r|}$ and rolling H_v is strong and positive at the full S&P 500 scale, with mean $r = 0.531$, 92.7% of 488 firms exhibiting positive coupling, and the phenomenon present across all eleven GICS sectors (H2 confirmed); (iii) the high-VIX regime exhibits substantially greater cross-firm dispersion of coupling and a higher median, with the Mann–Whitney test rejecting the null at $p = 0.0019$, so the regime amplification is in distribution shape, not mean shift (H3 confirmed in a sharper form than the original 50-firm sample suggested).

The firm-conditional predictive null is that the Coupling Intensity Index, a rolling Pearson correlation between two rolling Hurst series introduced here as a measurement object, has no firm-specific incremental predictive content for any of the ten forward outcomes tested in this paper under cluster-robust inference at the firm dimension (firm-clustered, two-way clustered, CR2, wild cluster restricted bootstrap). A subset of time-conditional methods returns focal-only significance, but that significance is not robust to the inference choice and does not survive horse-race conditioning. Every focal-only signal in a nine-target battery is fully absorbed by nine standard liquidity and volatility benchmarks in a horse-race specification (the cleanest case is the forward Corwin–Schultz spread, where focal two-way $t = +3.64$ collapses to combined $t = +0.07$). CII adds nothing to the existing forecasting toolkit at the firm-month unit of analysis.

The three methodological observations, summarised in Section 5.5.4 and Section 5.5.5, are: (a) the robust headline point, that share-volume vs. dollar-volume Amihud differ in a price-level multiplicative factor that any regressor correlated with within-firm price drift will appear to forecast, and cluster-robust inference does not detect this confound; (b) the result is sensitive to the inference dimension, since firm-conditional and time-conditional cluster-robust methods can disagree at the same data when residual dependence is strong on both dimensions, and the firm-conditional methods give the conservative reading that matches the firm-month unit of forecast this paper claims; (c) increasing cluster count from $G = 50$ to $G = 488$ did not raise the Bell–McCaffrey CR2 Satterthwaite effective degrees of freedom (it remained low, on the order of one in the focal specifications) because a small number of leverage-heterogeneous firms dominate the meat matrix, and the Satterthwaite check should be reported as a routine diagnostic rather than assumed to scale with G.

For practitioners the present results do not support using CII as a forward firm-specific liquidity indicator. Whether a cross-sectionally aggregated CII index has any value as a market-state indicator is not something the firm-month panel here can establish, and is left as an open question for a dedicated market-time-series study. For methodologists, the three observations above support using the standard Amihud (2002) dollar-volume convention, reporting cluster-robust inference at the dimension that matches the paper's claimed unit of analysis, and treating Satterthwaite df as a diagnostic rather than a guarantee.

Data and Code Availability

Daily OHLCV data are sourced from Yahoo Finance via the `yfinance` Python package. No proprietary or restricted data are used. All analysis code, the complete LaTeX manuscript source, and figure-generation scripts are archived at Zenodo (concept DOI: 10.5281/zenodo.19611543, always resolving to the latest version; this submission corresponds to v1.1.0, version DOI 10.5281/zenodo.19835451) and maintained at `https://github.com/mhdk1602/fractal-pv-coupling`. A step-by-step replication guide is provided in Appendix B. An interactive dashboard for exploring per-ticker results is available at `https://fractal-pv.streamlit.app`.

Declaration of Competing Interest

The author declares no competing interests.

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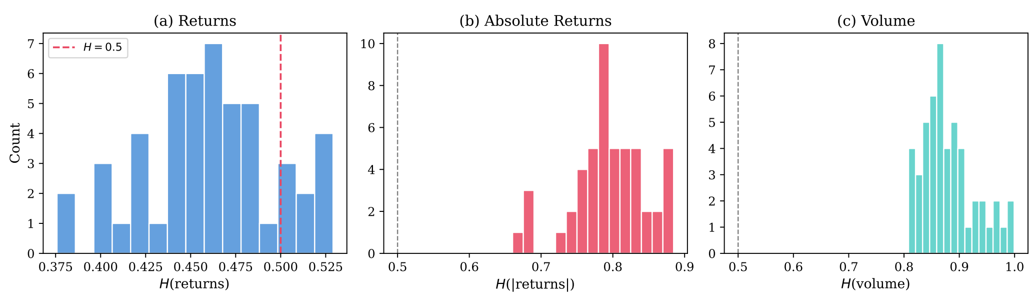

Figure 1: Distribution of full-sample Hurst exponents across 50 S&P 500 constituents (2015–2026). (a) Returns cluster around $H = 0.5$. (b) Absolute returns show strong long memory ($H \approx 0.8$). (c) Volume exhibits the strongest persistence ($H \approx 0.9$).

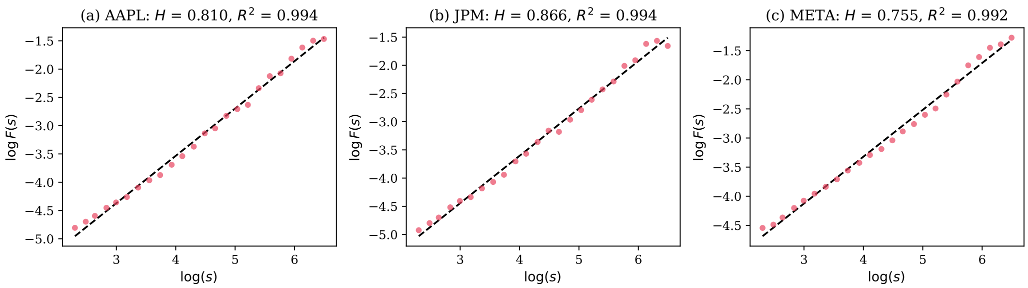

Figure 2: Log-log DFA plots for absolute returns (AAPL, JPM, META). High R^2 values confirm excellent linearity.

Figure 3: Rolling Hurst exponents for $H^{(i)}_{|r|,t}$ (red) and $H^{(i)}_{v,t}$ (teal). (a) AAPL: $r = 0.759$. (b) JPM: $r = 0.901$. (c) META: $r = -0.421$ (negative coupling during corporate restructuring).

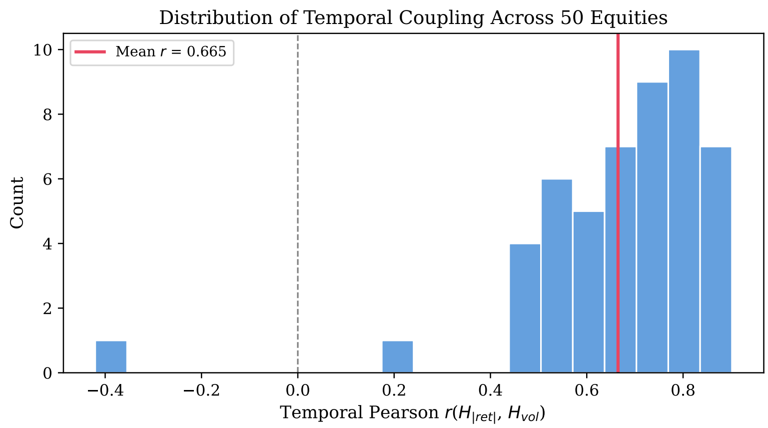

Figure 4: Distribution of temporal coupling across 50 equities. Mean $r = 0.665$; META is the sole outlier.

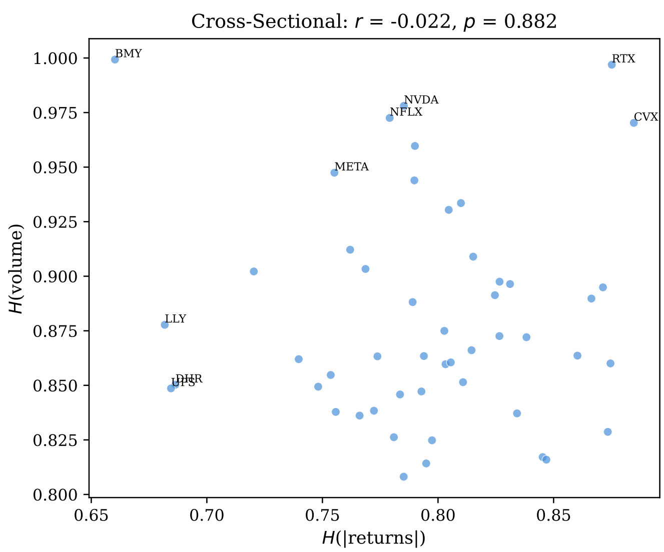

Figure 5: Cross-sectional scatter: $H^{(i)}_{|r|}$ vs. $H^{(i)}_v$. No relationship ($r = -0.02$, $p = 0.88$).

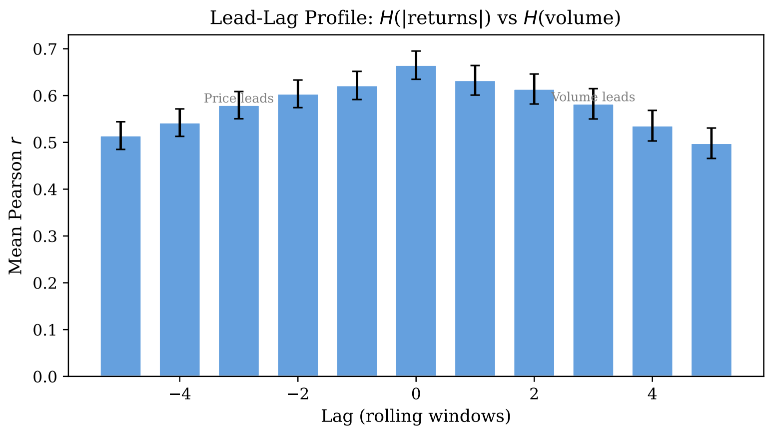

Figure 6: Average lead-lag cross-correlation profile across 50 equities. Lag zero is dominant; decay is symmetric.

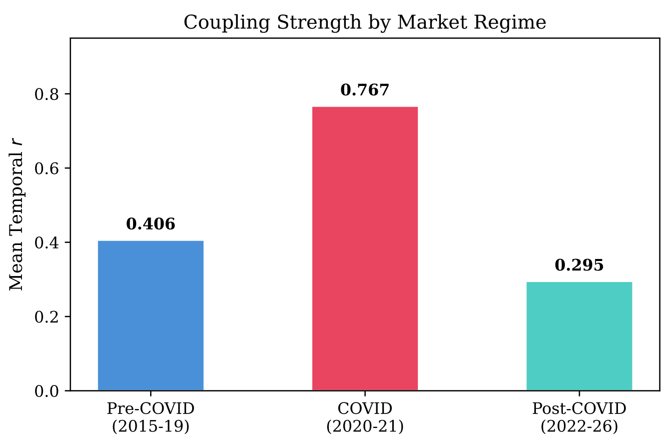

Figure 7: Temporal coupling by market subperiod. Coupling nearly doubles during COVID ($r = 0.77$) relative to the pre-COVID baseline ($r = 0.41$).

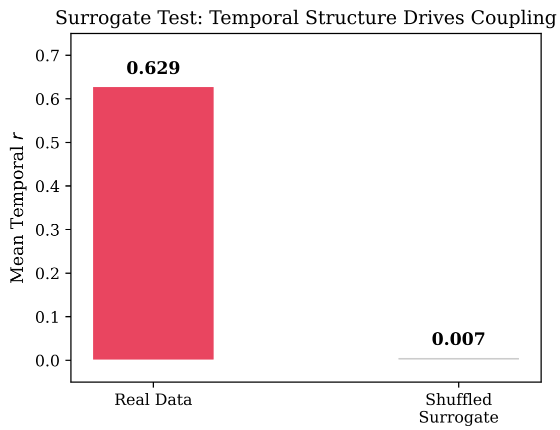

Figure 8: Surrogate test. Shuffling each series independently destroys coupling ($r = 0.629 \rightarrow 0.007$), confirming the temporal origin of the co-movement.

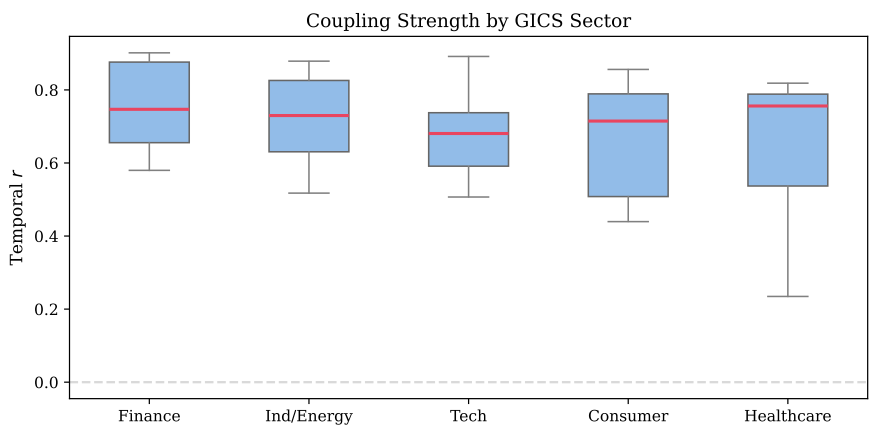

Figure 9: Temporal coupling by GICS sector at the original 50-firm sample (v1.0–v1.2). Present across all sectors; Finance strongest ($r = 0.756$). The G = 488 sector breakdown in Table A.9 reproduces the same qualitative pattern across all eleven GICS sectors, with Energy at the high end ($r = 0.66$) and Utilities at the low end ($r = 0.41$).

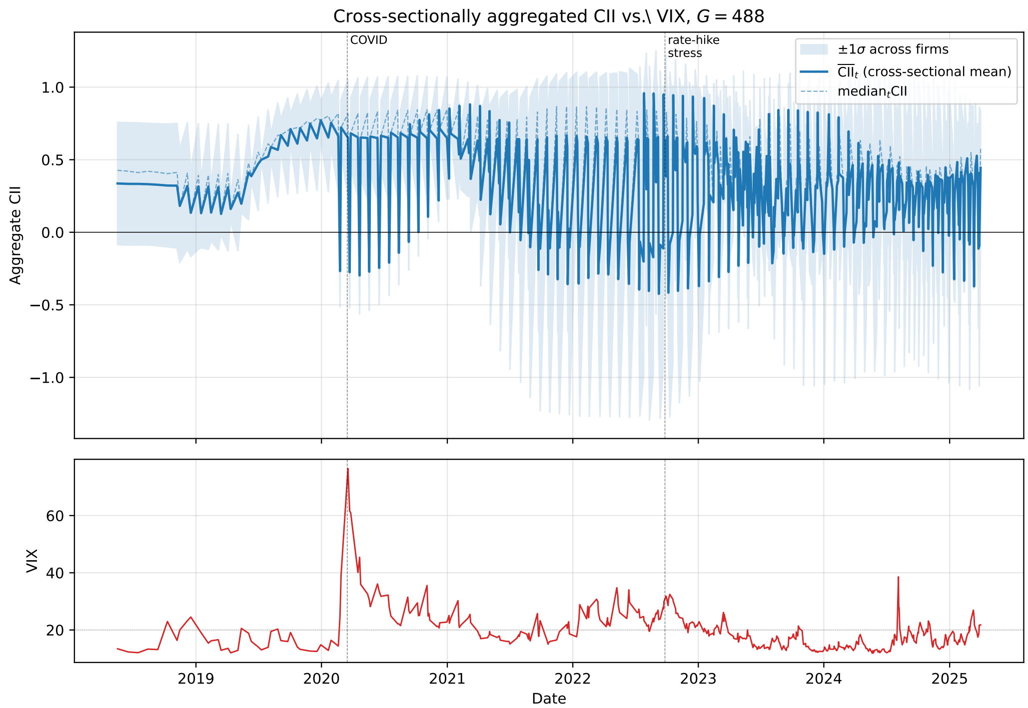

Figure 10: Cross-sectionally aggregated CII at $G = 488$ versus the CBOE VIX, 2018–2025. The top panel shows the cross-firm mean of CII at each rolling-Hurst evaluation date with a $\pm 1\sigma$ band reflecting cross-firm dispersion; the dashed line is the cross-firm median. The bottom panel shows the VIX. Aggregate CII covaries weakly but significantly with VIX: Pearson $r = 0.057$, Spearman $\rho = 0.097$ ($p = 0.033$). The cross-firm dispersion of CII rises sharply during stress episodes (COVID-19 in 2020, the 2022 rate-hike cycle), consistent with the H3 distribution-shape amplification reported in Table 1, but the mean of the cross-section is not strongly aligned with the VIX level. The focal-only significance that appears under time-conditional inference in Section 5.5 is therefore not even driven by a simple aggregate-CII-tracks-aggregate-stress channel, which reinforces reading it as sensitivity to the inference dimension rather than firm-specific predictive content; any market-time-series investigation of cross-firm dynamics is best left to a dedicated follow-up study at that unit of analysis.

Appendix A. Sample Composition

The Phase 2 panel comprises 488 firms drawn from the current April 2026 S&P 500 constituency. Of 503 attempted tickers, 4 were dropped at fetch for fewer than 600 daily observations over 2015–2026 (typically post-2023 IPOs), and approximately 11 were dropped during the panel build for stationarity or data-quality issues. The full constituent list is published with the replication materials at `https://github.com/mhdk1602/fractal-pv-coupling/blob/master/data/sp500_constituents_2026-04-28.csv`. Table A.9 summarises the GICS sector distribution of the 488-firm panel and the sector-level mean and median within-firm temporal coupling reported in Section 5.

Table A.9: Sector composition of the 488-firm Phase 2 panel, with within-firm temporal coupling statistics by sector.

GICS Sector	n firms	Mean r	% positive	Median r
Energy	22	0.660	95.5%	0.738
Consumer Discretionary	48	0.626	95.8%	0.726
Financials	76	0.610	94.7%	0.695
Health Care	56	0.543	94.6%	0.633
Consumer Staples	35	0.541	97.1%	0.622
Industrials	77	0.530	92.2%	0.616
Communication Services	23	0.476	87.0%	0.562
Information Technology	71	0.471	91.5%	0.547
Real Estate	31	0.461	87.1%	0.518
Materials	26	0.425	92.3%	0.486
Utilities	31	0.412	87.1%	0.451
Total / pooled	**488**	**0.531**	**92.7%**	**0.615**

The earlier 50-firm sample used in v1.0–v1.2 of this study was a large-cap-leaning subset of the same S&P 500 universe; the figures and robustness exhibits in this manuscript that depict the 50-firm sample are explicitly labelled as such in their captions. The G = 488 panel reproduces the H1, H2, and H3 results documented in v1.0–v1.2 at sharper precision, with the H3 result reformulated as distribution-shape amplification rather than mean amplification (see Section 5). The Phase 2 inclusion criteria, point-in-time membership convention, and survivorship-bias profile are documented in Section 4; the survivorship caveat is a feature of using current rather than CRSP-based historical constituency and is shared with most prior empirical work using the same construction.

Appendix B. Replication Guide

All code, data scripts, the pre-registration document, and the manuscript source are archived at Zenodo (concept DOI: 10.5281/zenodo.19611543, always resolving to the latest version) and on GitHub at `https://github.com/mhdk1602/fractal-pv-coupling`. The repository contains a Python package (`fractal_pv`) installable via `pip install -e .` with Python 3.10 or later. Dependencies are pinned in `pyproject.toml`; the conda-style environment can also be reproduced via `python -m venv .venv && .venv/bin/pip install -e .`.

Phase 2 ($G = 488$) replication.. The single canonical entry point is `research/horse_race/run_pha` which executes the seven-stage pipeline frozen under the pre-registration document `research/horse_race/PHASE2_PROTOCOL.md`: (1) batch fetch via `fractal_pv.data.fetch_universe_batch` from the constituent list at `data/sp500_constituents` (2) `fractal_pv.stationarity.prepare_series` for ADF/KPSS verification and log-return / log-volume construction; (3) `fractal_pv.rolling.rolling_dual_hurst` for aligned rolling DFA at $W = 500$ and step $\Delta = 20$; (4) `fractal_pv.predict.build_prediction` with $h = 21$ horizon and dollar-volume Amihud; (5) `fractal_pv.benchmarks.compute_all_bench` for the nine standard liquidity controls; (6) ten-target horse race via `fractal_pv.inference_robus` (7) per-target classification under the pre-registered decision rule. Per-benchmark regressions for the Table 4 cells are produced by the sibling script `research/horse_race/run_per_b` The respecification battery for log Amihud, ΔAmihud, forward Corwin–Schultz spread, forward Roll spread, and forward vol-of-vol is in `research/horse_race/respec_bat`

Earlier ($G = 50$) replication.. The original 50-firm pipeline used in v1.0–v1.2 of this study uses the sequential per-ticker fetcher `fractal_pv.data.fetch_universe` (preserved in the codebase) and the runner `research/horse_race/run_horse_race.py`. Both pipelines share the rolling DFA, predictive panel, benchmark, and inference machinery in `src/fractal_pv/`.

Inference machinery.. `fractal_pv.inference_robust.robust_panel_regression` produces six baseline standard-error specifications per regressor (HC1; firm-clustered; time-clustered; two-way clustered with the Cameron–Gelbach–Miller eigenvalue PSD adjustment; Newey–West; Driscoll–Kraay with Bartlett bandwidth ≥ 21). For the focal regressor, two further small-sample-corrected measures are computed: the Bell–McCaffrey CR2 standard error with Satterthwaite effective degrees of freedom (Imbens and Kolesár, 2016), and the wild cluster restricted bootstrap with Rademacher weights and restricted-null residuals (Cameron et al., 2008; MacKinnon and Webb, 2017).

Output artifacts.. Figures are generated via matplotlib and saved under `research/paper/figures/`. The manuscript compiles with `tectonic main.tex` or any standard LaTeX distribution. Intermediate panel and rolling-Hurst pickles are cached under `research/horse_race/cache/` for both pipelines; the cache layer ensures the seven-stage Phase 2 run is idempotent and that incremental re-runs reuse the slow rolling-DFA stage. Total wall-clock runtime for a clean Phase 2 run is approximately 60–75 minutes on a single-workstation configuration; the rolling-DFA stage parallelises across CPU cores via `joblib`.

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